

QM 2023 Capstone Project - Milestone 4 - Individual Contribution Statement Name: Steffi Brewer **Team:** 2nd row team **Date:** April 29th 2026

1. Personal Contribution to Capstone Milestones: **Milestone 1: Data Pipeline (Week 5) Tasks completed:**

Cleaned and standardised raw datasets for analysis & Helped produce a read me file, which included our research question and hypotheses. **Hours spent:** 5 hours **Key deliverable:** Readme file, which was updated with hypotheses and research direction. Datasets cleaned and produced so they were easy to visualise and interpret.

Milestone 2: EDA Dashboard (Week 10) Tasks completed: Helped to create key visualisations including correlation heatmap between returns, sentiment and controls & Wrote clear economic reports to showcase our key findings and hypotheses for M3. **Hours spent:** 5 hours. **Key deliverable:** Heatmap showing correlation between returns, sentiment and controls, and other data visualisations, M2 summary, including the hypotheses and key findings. **Milestone 3: Econometric Models (Week 14) Tasks completed:** Helped specific and interpret Fixed Effects regression model & Conducted robustness checks for example excluding COVID years, and different coefficients across firm size subsamples. **Hours spent:** 5 hours **Key deliverable:** M3 interpretation report.

Milestone 4: Final Investment Memo (Week 14) Tasks completed: Helped to draft executive summary & Supported my teammates in editing the memo ensuring clarity and professionalism. & Helped prepare my group for the final presentation through editing the power point file. **Hours spent:** 5 hours **Key deliverable:** Final business Memo. Final power point in preparation for our final Presentation. Total Estimated Contribution: **Total hours across all milestones:** 20 hours **Percentage of team workload:** 30% **Role(s) on team:** Communication lead, and team recorder for our findings and on our power point.

2. One Defended Methodological Decision. **Decision:** I supported using lagged macroeconomic variables in our Fixed Effects Regression model. I recommended using lagged sentiment variables in our Fixed Effects model. Exploratory analysis and robustness checks showed sentiment coefficients remained positive across alternative lag structures (baseline = 0.0277 vs 0.0078) relationship not driven by a single lag choice. Changes in consumer and investor sentiment take time to be reflected in market behaviour. M3 robustness checks confirmed stability of relationship excluding COVID years (coefficient = 0.0118)

3. One Key Limitation of Our Analysis. **Limitation:** Our Fixed Effects model assumes that unobserved REIT (property type exposure, leverage) characteristics are constant over time. In reality, REITs can change their portfolios over time, especially during periods like COVID. If these changes are correlated with both returns and macroeconomic exposure, our estimates may be biased. **Potential mitigation:** Future analysis could incorporate time varying controls or use a dynamic panel model to better capture changes overtime. Also, including sector time interaction effects could help control for industry wide shifts.

4. AI Audit Notes (If Applicable)

AI Tools Used:

Our use of AI was fully documented in each Milestone AI Audit Appendix.

5. Self-Reflection

What did I do particularly well on this capstone? I excelled at translating our research on consumer and investor sentiment into clear meaningful insights into stock market behaviour. I helped connect our core research data to our research question ensuring that our research was focused and results driven. What could I have improved? I could have dived deeper into our research to give me a broader understanding of its impact on the stock market and therefore I would have had more time to explore alternative research data and deeper robustness checks on the relation between sentiment and market return.

What did I learn from this capstone project? This capstone project taught me how complex and important data cleaning and alignment are when working with multiple sources like financial datasets and sentiment surveys. I learned how to interpret and evaluate whether sentiment acts as a predictor or contrarian indicator using economic reasoning and evidence.

6. Attestation **By submitting this individual addendum, I affirm that:** All contributions listed above are accurate and honest. I have not exaggerated my role or minimized teammates' contributions. I understand that this addendum may be used to adjust my individual grade relative to the team grade. I take full responsibility for my work and any errors in the sections I authored. **Signature:** Steffi Brewer **Date:** April 29th 2026